Proof Verification Dossier

Bounds and Estimators for the Civilian-to-Combatant Death Ratio

Internal document for manual checking — not for submission.

Machine companions: `analysis/verify_proofs.py` (sympy + Monte Carlo, all checks pass) and `formal/casualty_proofs` (Lean 4 + mathlib, kernel-checked, no `sorry`s).

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How to read this document. Each claim is stated exactly as in the manuscript, followed by a proof written in more detail than the paper's appendix, an explicit list of the assumptions used at each step, and a verification note describing the symbolic (sympy) or Monte-Carlo check performed. Most content is additionally machine-checked in Lean 4 with mathlib (`formal/casualty_proofs`): Claims 1–6 in full (identification algebra, monotonicity and derivative, η-slack, manpower bound, delta-method gradients, Gauss–Markov optimality), the infimum and monotonicity ingredients of Claim 7, the two-point contamination kernel of Claim 8, and the Section 6 arithmetic over exact rationals. The Lean development compiles with no unproven statements; only the fully measure-theoretic parts (posterior quantiles, the topological limit step) rest on the symbolic and Monte-Carlo checks alone.

Contents

1 Setup and standing conventions

Population of size N; deaths D in the window $[t_0, t_1]$. Classes: W (women of all ages + males under 18) with population share w, and AM (*all* males 18+) with population share a, so

$$w + a = 1 \quad \text{exactly.}$$

The exact partition matters: the identification formulas substitute $a = 1 - w$, so any convention in which AM excludes part of the male population (e.g. men over 60) silently breaks them.

Combatant stock M, $f = M/N$, all combatants in AM (slack for exceptions handled in Claim 3). Estimand $q = D_M/D$, the combatant share of the dead; ω = share of the *dead* in class W.

The hazard model. All identification results use one structural model: conditional on being a civilian, a member of class W dies with per-capita hazard λ and a member of civilian-AM with hazard $\mu\lambda$, $\mu \geq 1$. Combatant deaths are governed separately (their total is qD by definition). No assumption is made about λ itself — it cancels.

2 Claim 1: benchmark identification (paper eq. 2, Assumption 3.1)

Claim 1 (Point identification under exchangeable collateral)**.** *If all civilians face equal per-capita hazard ($\mu = 1$), then*

$$q = 1 - \frac{\omega(1-f)}{w}.$$

Proof. Civilian population masses: class W has wN civilians (no combatants in W); civilian AM has $(a-f)N$. With equal hazard λ, expected civilian deaths in W and civilian-AM are λwN and $\lambda(a-f)N$, so the fraction of *civilian* deaths landing in W is

$$\frac{wN}{wN + (a-f)N} = \frac{w}{w + a - f} = \frac{w}{1-f},$$

where the last step uses $w + a = 1$. Combatant deaths (share q of all deaths) land entirely in AM. Hence the share of *all* deaths in W is

$$\omega = (1-q) \cdot \frac{w}{1-f},$$

and solving for q gives the display. Note this is exact in expectation (a first-moment identity), not an approximation; finite-sample noise is handled by the delta method (Claim 5). □

Verification status. **VERIFIED**. Sympy: solving $\omega = (1-q)w\lambda/(w\lambda + (a-f)\mu\lambda)$ for q and substituting $\mu = 1$, $a = 1 - w$ reproduces the display exactly (λ cancels symbolically). Lean: `claim1_benchmark_identification`. Degenerate cases: $\omega = w$, $f = 0 \Rightarrow q = 0$; $\omega = 0 \Rightarrow q = 1$. *Subtlety for manual checking:* the last step of the proof needs $w + a = 1$ exactly (see the convention note in Section 1); with $a < 1 - w$ the two displayed forms of the identity disagree.

3 Claim 2: sharp identified set (paper Theorem 3.3)

Claim 2 (Sharp identified set under μ-bounded exposure)**.** *Suppose $f \leq a$ and the hazard model holds for some $\mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$. Given (ω, w, a, f),*

$$q(\mu) = 1 - \omega\,\frac{w + \mu(a-f)}{w},$$

is monotone decreasing in μ, and the identified set for q is exactly $[\,q(\mu_{\text{hi}}),\ q(\mu_{\text{lo}})\,] \cap [0, 1]$.

Proof. Step 1 (the curve). With hazards $(\lambda, \mu\lambda)$, expected civilian deaths are proportional to $\lambda wN + \mu\lambda(a-f)N$, so the civilian share landing in W is $w/(w+\mu(a-f))$ and

$$\omega = (1-q)\,\frac{w}{w+\mu(a-f)}. \tag{1}$$

Solving (1) for q gives $q(\mu)$.

Step 2 (monotonicity). $\partial q/\partial\mu = -\omega(a-f)/w \le 0$, with equality iff $a = f$ (no civilian adult men) or $\omega = 0$.

Step 3 (validity: every point of the interval with $q(\mu^*) \in [0,1]$ is attainable). Fix any $\mu^* \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$ such that $q(\mu^*) \in (0,1)$ (values outside $[0,1]$ are not shares and are excluded by the final intersection), and any $\lambda > 0$ small enough that all implied class death counts are below class populations. The pair $(\lambda, \mu^*\lambda)$ is an admissible hazard configuration whose implied ω satisfies (1) with $q = q(\mu^*)$. Boundary points are obtained in the closure: $q(\mu^*) = 1$ corresponds to $\omega = 0$ (no civilian deaths), attained by $\lambda = 0$ or as the limit $\lambda \downarrow 0$; $q(\mu^*) = 0$ is attained directly with $\lambda > 0$ and no combatant deaths. Hence $q(\mu^*)$ is consistent with the data (ω, w, a, f): the identified set contains $\{q(\mu) : \mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]\} \cap [0,1]$, which by continuity and monotonicity is $[q(\mu_{\text{hi}}), q(\mu_{\text{lo}})] \cap [0,1]$.

Step 4 (sharpness: nothing outside is attainable). Conversely, any admissible configuration has some $\mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$, and (1) then forces $q = q(\mu)$, which lies in the interval. The intersection with $[0,1]$ imposes the logical range of a share. □

Verification status. **VERIFIED**. Sympy confirms the inversion and the derivative; Lean: `claim2_inversion`, `claim2_monotone`. 2,000 random parameter draws confirm endpoint ordering. The forward model's three death-shares (combatant, civilian-AM, W) sum to 1 symbolically. *Caveat worth knowing when checking manually:* the theorem treats ω as an exact population quantity; the "identified set" is a set for the estimand given error-free inputs. Sampling error enters separately via Claim 5 and systematic error via the anchor sensitivity analysis. This is stated in the paper's Discussion (limitation 3).

4 Claim 3: combatants outside AM (paper Remark 3)

Claim 3 (η-slack)**.** *Suppose a fraction $\eta \in [0, \bar\eta]$, $\bar\eta = 0.03$, of combatant* deaths *is recorded in class W, while the combatant stock stays in* AM *(a death-classification slack, so $S := w/(w+\mu(a-f))$ is unchanged). The observable identity becomes $\omega = (1-q)S + \eta q$, so*

$$q_{\text{true}} = \frac{S-\omega}{S-\eta}, \qquad q_{\text{true}} - q_{\eta=0} \;=\; \frac{\eta(S-\omega)}{S(S-\eta)} \;=\; \frac{\eta\, q_{\text{true}}}{S} \;=\; \frac{\eta\, q_{\eta=0}}{S-\eta},$$

and the bias is ≥ 0 whenever $S > 0$ and $q_{\text{true}} \ge 0$: ignoring the slack understates *q. Provided $S > \bar\eta$ (which holds at the Gaza values, where $S \ge 0.597$ for $\mu \le 2$), a loose uniform bound over $\mu \in [1,2]$ with $q_{\eta=0} \le 0.26$ is $\bar\eta \cdot 0.26/(0.597 - \bar\eta) \approx 1.4$ percentage points; the exact per-μ bias is smaller and decreasing in μ (1.05 points at $\mu = 1$, 0.74 at $\mu = 1.5$, 0.33 at $\mu = 2$), because $q_{\eta=0}$ falls with μ faster than S does.*

Proof. Deaths in W now come from two sources: civilian deaths landing in W, which is $(1-q)S$ of all deaths as in Claim 2, plus the fraction η of combatant deaths, contributing ηq. Setting $\omega = (1-q)S + \eta q = S - q(S-\eta)$ and solving for q gives $q_{\text{true}} = (S-\omega)/(S-\eta)$. Subtracting $q_{\eta=0} = (S-\omega)/S$,

$$q_{\text{true}} - q_{\eta=0} = (S-\omega)\Big(\frac{1}{S-\eta} - \frac{1}{S}\Big) = \frac{\eta(S-\omega)}{S(S-\eta)}.$$

Substituting $S-\omega = q_{\text{true}}(S-\eta)$ gives the second form $\eta q_{\text{true}}/S$; substituting $S-\omega = q_{\eta=0}S$ gives the third form $\eta q_{\eta=0}/(S-\eta)$.

Nonnegativity. From the second form, $q_{\text{true}} - q_{\eta=0} = \eta q_{\text{true}}/S \geq 0$ for $S > 0$, $\eta \geq 0$, $q_{\text{true}} \geq 0$. (This is the right route: arguing instead from "$\omega \leq S$" requires $S \geq \eta$, since $\omega = S - q(S-\eta)$ gives $\omega \geq S$ when $\eta > S$ and $q \geq 0$.)

Bound. For the uniform bound the third form is used with the worst admissible values: it is increasing in η (for $S > \eta$) and in $q_{\eta=0}$, and decreasing in S, so on $\eta \leq \bar{\eta}$, $q_{\eta=0} \leq 0.26$, $S \geq 0.597$ (the $\mu = 2$ value; S decreases in μ) it is at most $\bar{\eta} \cdot 0.26/(0.597 - \bar{\eta}) \approx 0.0138$. This is loose because it pairs the smallest S (attained at $\mu = 2$) with the largest $q_{\eta=0}$ (attained at $\mu = 1$): the exact per-μ bias $\bar{\eta}\, q_{\eta=0}(\mu)/(S(\mu) - \bar{\eta})$ evaluates to 0.0105, 0.0074, 0.0033 at $\mu = 1, 1.5, 2$.

Note the distinction between the second and third forms: the bias equals $\eta q/(S-\eta)$ with q the *no-slack* estimate, and $\eta q/S$ with q the *true* share — the two coincide only to first order in η. □

Verification status. **VERIFIED**. Sympy: all three displayed forms are symbolically equal to $q_{\text{true}} - q_{\eta=0}$; the superficially plausible chain $\eta q_{\text{true}}/S \cdot S/(S-\eta)$ (which double-counts the $(S-\eta)^{-1}$ correction) is confirmed *not* to equal the bias. Lean: the identity `claim3_qtrue`, the three bias forms, nonnegativity exactly under $S > 0$, $\eta \geq 0$, $q_{\text{true}} \geq 0$ (theorem `claim3_bias_nonneg_of_qtrue`), the spurious-chain refutation, and exact rational bounds on the per-μ biases. Numeric: $S(1) = 0.748$, $S(1.5) = 0.664$, $S(2) = 0.597$ at Gaza demographics — so any blanket support claim like "$S \geq 0.70$" would fail for $\mu \gtrsim 1.15$, which is why the claim states the condition as $S > \bar{\eta}$ and reports per-μ biases. Direction note: the slack makes the paper's *upper* bounds conservative in the claim-favourable direction, so no downstream conclusion depends on suppressing it.

5 Claim 4: manpower bound (paper Corollary 3.4)

Claim 4. *$q \leq M/D$; whenever $\max\{0, q(\mu_{\text{hi}})\} \leq \min\{1,\, q(\mu_{\text{lo}}),\, M/D\}$, the joint identified set is $\big[\max\{0, q(\mu_{\text{hi}})\},\ \min\{1,\, q(\mu_{\text{lo}}),\, M/D\}\big]$; otherwise it is empty and the inputs are jointly infeasible.*

Proof. Combatants killed cannot exceed combatants existing: $D_M \leq M$. Divide by $D > 0$. The joint set is the intersection of two intervals, hence the endpoint formulas. (If the intersection is empty, the inputs are jointly infeasible — this is exactly the event detected by a positive contradiction radius.) □

Verification status. **VERIFIED** (algebraic identity; Lean: `claim4_manpower_bound`). One caveat for manual review: M here is the stock at t_0; if recruitment during the war is material, M should be the stock-plus-inflow, which *raises* the bound. In the Gaza application the bound is slack by a factor > 2 ($M/D = 0.64$ vs. $q \leq 0.26$), so recruitment cannot change any conclusion.

6 Claim 5: delta-method standard error (paper §3)

Claim 5. *For $\hat{q} = 1 - \hat{\omega}(1-\hat{f})/\hat{w}$ with independent inputs,*

$$\widehat{\text{Var}}(\hat{q}) \approx \frac{\hat{\omega}^2}{\hat{w}^2}\widehat{\text{Var}}(\hat{f}) + \frac{(1-\hat{f})^2}{\hat{w}^2}\widehat{\text{Var}}(\hat{\omega}) + \frac{\hat{\omega}^2(1-\hat{f})^2}{\hat{w}^4}\widehat{\text{Var}}(\hat{w}).$$

Proof. First-order Taylor expansion of $g(\omega, f, w) = 1 - \omega(1-f)/w$:

$$\partial_f g = \frac{\omega}{w}, \qquad \partial_\omega g = -\frac{1-f}{w}, \qquad \partial_w g = \frac{\omega(1-f)}{w^2}.$$

$\mathrm{Var}(\hat{q}) \approx \sum (\partial g)^2 \mathrm{Var}$, using independence (no covariance terms). Squaring the gradients gives the display. □

Verification status. **VERIFIED**. All three partial derivatives and the assembled formula verified symbolically; the gradients are also kernel-checked as `HasDerivAt` statements in Lean (`claim5_grad_f/om/w`). Assumption to flag for manual review: independence of $(\hat{\omega}, \hat{f}, \hat{w})$ is plausible here because they come from different institutions (deaths records, military-balance estimates, census), but any common-mode error (e.g. both ω and w derived from the same census frame) would add covariance terms. The paper's conservative $\sigma_\omega = 0.01$ ($5\times$ the binomial SE) partially absorbs this.

7 Claim 6: precision-weighted aggregation (paper Proposition 2.1)

Claim 6. *For independent unbiased sources with known variances σ_i^2 (with estimated $\hat{\sigma}_i^2$ independent of the point estimates the plug-in combination remains conditionally unbiased but not in general minimum-variance in finite samples; when the estimates are also consistent it is asymptotically efficient among linear unbiased combinations, under standard regularity), the estimator*

$$\hat{\theta}_{\mathrm{agg}} = \frac{\sum_i \hat{\theta}_i/\hat{\sigma}_i^2}{\sum_i 1/\hat{\sigma}_i^2}$$

is minimum-variance among linear unbiased combinations with weights fixed given the variances. With per-source biases b_i, the aggregate bias is the precision-weighted mean of the b_i. Under ε_i-contamination in which the contaminated report is displaced from the truth by at most R_i (equivalently, the contaminating support is contained in a set of diameter R_i containing the truth), the worst-case aggregate bias is at most $\sum_i \varepsilon_i R_i \, \hat{\sigma}^2_{\mathrm{agg}}/\hat{\sigma}_i^2$.

Proof. (i) Minimise $\sum c_i^2 \hat{\sigma}_i^2$ subject to $\sum c_i = 1$. The Lagrangian condition gives $2c_i\hat{\sigma}_i^2 = \kappa$, i.e. $c_i \propto 1/\hat{\sigma}_i^2$; the constraint fixes the normalisation. The objective is strictly convex, so this is the unique minimum (Gauss–Markov in one dimension).

(ii) $\mathbb{E}\hat{\theta}_{\mathrm{agg}} - \theta = \sum_i c_i b_i$ with $c_i = \hat{\sigma}^2_{\mathrm{agg}}/\hat{\sigma}_i^2$; direct substitution.

(iii) Contamination replaces source i's report with probability ε_i by a value displaced by at most R_i (diameter of the contaminating support). The expected displacement of source i's contribution is at most $\varepsilon_i R_i$, weighted into the aggregate by c_i; summing over i gives the bound. □

Verification status. **VERIFIED**. Optimality confirmed on 300 random configurations against random competitor weights; bias formula confirmed symbolically. In Lean, the precision weights are proved to sum to one and the minimum-variance inequality $\sum c_i^2\sigma_i^2 \geq 1/\sum \sigma_i^{-2}$ is proved for every unbiased combination via Cauchy–Schwarz (`claim6_optimality`). Nuance for manual review: in (iii) "diameter of the support" means the contaminated value cannot leave a set of diameter R_i around truth; if the adversary is unbounded, so is the worst case — this is why the paper pairs the proposition with the identified-set diameters R_i (removal diameters), which are finite by construction. The bound is on *expected* bias, not worst-case realised displacement.

8 Claim 7: feasibility and contradiction (paper Theorem 4.3)

Claim 7. *Fix* $(\hat{a}, \hat{f}, \hat{D})$ *and* $[\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$. *Define* $\mathcal{F}(x)$ *as in the paper (eq. 4) and*

$$\rho^{(i)} = \inf_{\omega', w', M'} \max\Big\{\tfrac{|\omega'-\hat{\omega}|}{\sigma_\omega}, \tfrac{|w'-\hat{w}|}{\sigma_w}, \tfrac{(M'-\hat{M})_+}{\sigma_M}\Big\} \text{ s.t. } q^{(i)} \in \mathcal{F}((\omega', w', \hat{a}, \hat{f}, M', \hat{D})).$$

Then (i) $\rho^{(i)} = 0$ *iff* $q^{(i)} \in \mathcal{F}(\hat{x})$*, and (ii) if* $\rho^{(i)} > 0$*, every rationalisation* (ω', w', M') *has at least one of* $|\omega' - \hat{\omega}|/\sigma_\omega$*,* $|w' - \hat{w}|/\sigma_w$*,* $(M' - \hat{M})_+/\sigma_M$ *at least* $\rho^{(i)}$*; that is, at least one input must move by* $\rho^{(i)}$ *standard errors* in a penalised direction *(downward moves of* M *are free by construction, since lowering the manpower ceiling can only hurt the claim).*

Proof. (i, $\Leftarrow$*)* If $q^{(i)} \in \mathcal{F}(\hat{x})$, take $(\omega', w', M') = (\hat{\omega}, \hat{w}, \hat{M})$: objective 0.

(i, $\Rightarrow$*)* Suppose $\rho^{(i)} = 0$. Take a sequence (ω'_n, w'_n, M'_n) of feasible points with objective $\to 0$. Because the M-penalty is *one-sided*, this does *not* imply $M'_n \to \hat{M}$; it implies only

$$\omega'_n \to \hat{\omega}, \qquad w'_n \to \hat{w}, \qquad (M'_n - \hat{M})_+ \to 0,$$

i.e. $\limsup_n M'_n \le \hat{M}$ (with M'_n possibly far below $\hat{M}$). We therefore argue via the constraint rather than via convergence of M'_n. Each feasible point satisfies (a) $q^{(i)} = 1 - \omega'_n(w'_n + \mu_n(a - f))/w'_n$ for some $\mu_n \in [\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$, and (b) $q^{(i)} D \le M'_n$. For (a): pass to a subsequence with $\mu_n \to \mu^*$ in the compact interval; continuity of $(\omega', w', \mu) \mapsto 1 - \omega'(w' + \mu(a - f))/w'$ (on $w' > 0$) gives $q^{(i)} = 1 - \hat{\omega}(\hat{w} + \mu^*(a - f))/\hat{w}$, so the exposure constraint holds at the observed $(\hat{\omega}, \hat{w})$. For (b): $q^{(i)} D \le M'_n \le \hat{M} + (M'_n - \hat{M})_+ \to \hat{M}$, so $q^{(i)} D \le \hat{M}$. Hence $q^{(i)} \in \mathcal{F}(\hat{x})$. (The one-sided penalty is why the argument must run through the constraint: "objective $\to 0$ implies the full vector converges to $(\hat{\omega}, \hat{w}, \hat{M})$" is false when downward moves of M are free; only the inequality $q^{(i)} D \le \hat{M}$ survives the limit, and that is all the theorem needs.)

(ii) Let (ω', w', M') be any point of C ("rationalisation"). By definition of the infimum, $\max\{|\omega' - \hat{\omega}|/\sigma_\omega, |w' - \hat{w}|/\sigma_w, (M' - \hat{M})_+/\sigma_M\} \ge \rho^{(i)}$, so at least one of the three *penalised* displacements is $\ge \rho^{(i)}$ SEs. This does not assert "M moved" generically: lowering M' costs nothing under the objective (and only shrinks the feasible set), so a positive radius can only ever be certified by an upward move of M, or a move of ω or w in either direction. ☐

Verification status. **VERIFIED**. Kernel-checked ingredients (Lean): the M-monotonicity that makes the one-sided penalty sound (`claim7_feasibility_monotone_in_M`), the antitonicity of ω_{needed} in μ behind the code's radius reduction, and the abstract infimum facts for parts (i,$\Leftarrow$) and (ii) — the radius lower-bounds every rationalisation's penalty and vanishes at a feasible face-value point. The limit step of (i,$\Rightarrow$) is verified numerically: 2,000 random infeasible claims all fail to become feasible under 10^{-9}-perturbations (no boundary pathologies found), and 500 simulated feasible sequences with M'_n far below $\hat{M}$ confirm that only $q^{(i)} D \le \hat{M}$ survives the limit. The reduction used in the code (radius on the ω-axis = distance from $\hat{\omega}$ to the interval $[\omega_{\mathrm{needed}}(\mu_{\mathrm{hi}}), \omega_{\mathrm{needed}}(\mu_{\mathrm{lo}})]$) is justified by ω_{needed} decreasing in μ (verified symbolically). Note for manual checking: Definition 4.2 grants the claim the most favourable μ in the admissible range, so the radius must be evaluated at that μ, not at the analyst's preferred value; `analysis/validate_bounds.py` enforces this.

9 Claim 8: bias-robust sensitivity band (paper Proposition 5.1)

Claim 8. *Let* $I_0 = [L_0, U_0]$ *be the* $1 - \alpha$ *credible interval under face-value sources, posterior supported on the identified set implied by the sources used. Each source* i *is independently contaminated with probability* ε_i*; removing the sources in* T *leaves an identified set of diameter* R_T *(*$R_i := R_{\{i\}}$*). Then: (i) conditional on realised pattern* T*, every* support-respecting $1 - \alpha$ *credible interval — one whose endpoints lie in the posterior support, e.g. equal-tailed or HPD*

— *lies within* $[L_0 - R_T,\, U_0 + R_T]$*; (ii) the* expected *endpoint displacement is at most* $\sum_i \varepsilon_i R_i$ *plus* $O(\varepsilon^2)$ *terms involving multi-removal diameters, so* $I_\varepsilon = [L_0 - \sum_i \varepsilon_i R_i,\ U_0 + \sum_i \varepsilon_i R_i]$ *is a first-order sensitivity band,* not *a uniform envelope over realised intervals; (iii) if the posterior mixture weight on contaminated components is at most* $\tau := 1 - \prod_i (1 - \varepsilon_i)$ *and the face-value posterior density exceeds* $m > 0$ *on the segment between the clean and contaminated quantiles,*

$$|\hat{Q}_\varepsilon(u) - \hat{Q}_0(u)| \ \le\ m^{-1}\tau, \qquad u \in \{\alpha/2,\, 1 - \alpha/2\}.$$

Proof. (i) Condition on T: honest sources outside T confine θ to the removal identified set of diameter R_T, which contains the face-value posterior support — in particular both L_0 and U_0. Any posterior formed under adversarial reports from T is supported there, so each endpoint of a support-respecting credible interval is within R_T of the corresponding face-value endpoint. The support restriction is necessary: a mathematically arbitrary $1 - \alpha$ credible interval can be widened without limit while retaining $1 - \alpha$ coverage, so no finite envelope can contain *every* credible interval.

(ii) The pattern probability is $P(T) = \prod_{i \in T} \varepsilon_i \prod_{i \notin T} (1 - \varepsilon_i)$, so the expected displacement is at most $\sum_T P(T) R_T = \sum_i \varepsilon_i R_i \prod_{j \ne i} (1 - \varepsilon_j) + \sum_{|T| \ge 2} P(T) R_T \le \sum_i \varepsilon_i R_i + O(\varepsilon^2)$. This is a statement about the *mean* over the contamination pattern. A quantile (interval endpoint) of the realised posterior can move by up to R_T on a contamination event of probability ε_i: with one source, $\varepsilon = 0.04$, $R = 100$, a contaminated component at distance $\approx R$ carrying 4% posterior mass moves the 97.5% quantile by $\approx R$, far beyond $\varepsilon R = 4$. Hence the band is first-order in ε, exact when at most one source can be contaminated *and* displacement is measured in expectation. Note also that R_T for $|T| \ge 2$ is not bounded by $\sum_{i \in T} R_i$ in general: two mutually corroborating sources can each have a small removal diameter while their joint removal is much larger.

(iii) Write $\hat{F}_\varepsilon = (1 - \pi)\hat{F}_0 + \pi G$ with $\pi \le \tau$; since CDFs are $[0, 1]$-valued, $\sup_x |\hat{F}_\varepsilon(x) - \hat{F}_0(x)| \le \pi \le \tau$, also at left limits. With quantiles as generalised inverses, $\hat{F}_\varepsilon(Q_\varepsilon^-) \le u \le \hat{F}_\varepsilon(Q_\varepsilon)$ and $\hat{F}_0(Q_0) = u$ ($\hat{F}_0$ is continuous where its density is positive), so $\hat{F}_0(Q_\varepsilon) \in [u - \tau, u + \tau]$, i.e. $|\hat{F}_0(Q_\varepsilon) - \hat{F}_0(Q_0)| \le \tau$; the density floor m on the segment between the two quantiles lower-bounds the left side by $m|Q_\varepsilon - Q_0|$, giving the display. The weight condition does *not* follow from the prior probabilities alone: posterior component weights are $\propto P(T)\, Z_T$ where Z_T is the component marginal likelihood, so an adversarial component that fits the data better than the clean model carries posterior weight exceeding τ. □

Verification status. **VERIFIED** (as stated; the statement is deliberately *not* a uniform credible-interval envelope, and the proof of (ii) exhibits why no such envelope exists for tail quantiles). The kernel of that distinction is machine-checked in Lean on a two-point contamination distribution (mass ε at R): the mean displacement equals εR exactly while the CDF stays $\le 1 - \varepsilon$ below R, so any quantile above level $1 - \varepsilon$ moves by the full R (`claim8_mean_displacement`, `claim8_quantile_stuck_below`). Numerics: the one-source configuration $\varepsilon = 0.04$, $R = 100$, clean posterior $\mathcal{N}(0, 0.1^2)$, contaminated component $\mathcal{N}(100, 0.1^2)$ moves the pooled 97.5% quantile to 99.97 while $\varepsilon R = 4$, confirming that only the expectation — not the realised quantile — obeys the first-order band. The exchange-of-sums identity behind (ii) is verified by exact enumeration over contamination patterns (100 random configurations), and the quantile bound in (iii) is confirmed on 200 Gaussian-mixture configurations where the weight condition holds by construction. Downstream use in Section 6: the quoted 32.1% → 39.4% inflation is the first-order band I_ε and is labelled as such; the paper's headline rejections rest on the bounds and the contradiction radius, not on this proposition.

10 Section 6 arithmetic (spot summary)

All application numbers are generated by `analysis/validate_bounds.py`. Key identities re-verified here at $(w, a, f, \omega) = (0.755, 0.245, 0.020, 0.523)$; the starred rows are also proved over exact rationals in Lean.

Quantity	Value	Check
$q(1)$	32.11%	symbolic + numeric + Lean* (exact: 12123/37750)
$q(1.5)$	24.32%	numeric + Lean*
$q(2)$	16.53%	numeric + Lean*
root of $q(\mu) = 0$	$\mu^* = 3.06$	numeric; matches the q=0 symmetry discussion
μ needed for $q = 17{,}000/71{,}444$	1.53	inversion; ≥ 1 so arithmetically feasible
μ needed for $q = 25{,}000/71{,}444$	0.82	inversion; < 1 so infeasible for any admissible μ
M/D	0.630	non-binding vs. all bounds above
η-slack bias, μ=$1, 1.5, 2$	1.30, 1.10, 0.83 pp	numeric + Lean*; uniform bound 1.66 pp